



Project report

Fragments of Growth

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Course Name: Computer Graphics

Course Number: CS2206

Theoretical Group: 2

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Task Table:

Scene	Member Name
1- cover	Safaa, Diala
2- Forest	Diala
3- Desert	Shatha
4- Beach	Dana
5- City	Abrar
6- Library	Safaa
7- Garden	Noor
8- End	Safaa
Report	Entire team
PPT	Entire team
Code organization	Entire team
Project organization	Safaa Ahmed Sobh



Chapter 1: Introduction

1.1 Abstract

Fragments of Hope is an interactive 2D computer graphics project that merges storytelling with game mechanics. The narrative follows a character through a desaturated, grayscale world that gradually regains its vibrancy as the player collects symbolic "Fragments of Hope". Each fragment represents a core human value: Identity, Courage, Clarity, Connection, and Purpose. By implementing various transformation and interaction techniques, the project demonstrates the practical application of the OpenGL pipeline in creating an engaging, multi-level user experience.

1.2 Introduction

The project scope encompasses six main distinct scenes, ranging from a desolate forest to a complex library. The primary goal was to create a cohesive narrative where the visual environment responds dynamically to player progression. Users navigate these environments using a combination of keyboard and mouse interactions, moving from introductory story elements to functional gameplay puzzles.



Chapter 2: Methodology

2.1 Functionality & Implementation:

The technical core of the project relies on the following OpenGL implementations:

- **Geometric Transformations:** We implemented `glTranslatef` to handle smooth, coordinate-based character locomotion across various 2D environments. To bring the world to life, we utilized `glRotatef` for dynamic animations, such as the continuous rotation of collectible `fragments` and the "falling" sequence used as a feedback mechanism in the desert `lose-state`. Additionally, `glScalef` was applied to create a "pulsing" effect, subtly altering the size of interactable items to draw the player's attention. (Nagoor, 2026)
- **Primitive Modeling:** To move beyond simplistic shapes, we utilized `GL_TRIANGLE_FAN` to render high-fidelity `circles for character features` and celestial objects like the `sun and moon`. Environmental assets, including the modular trees and desert mountain ranges, were constructed through a strategic combination of `GL_POLYGON` and `GL_TRIANGLES`, ensuring a visually cohesive and non-simplistic aesthetic throughout the narrative. (Nagoor, 2026)
- **Interaction Systems:** Navigation is facilitated through `glutSpecialFunc`, providing a precise response to arrow-key inputs. For the "Purpose" puzzle in the Library scene, we implemented a sophisticated mouse-interaction system using `glutMouseFunc` and `glutMotionFunc`. This allows users to physically drag and reorder book objects, translating real-time screen coordinates into world-space movement to solve the objective. (Nagoor, 2026)
- **Texture Mapping:** To elevate the visual quality, we developed a custom BMP loader that maps 2D images onto quads, providing professional "Cover" and "End" screens. We also simulated 2D atmospheric lighting by using `GL_BLEND` and `glBlendFunc` to create semi-transparent halos around active nodes and manually rendered gradient shadows in the library to enhance the sense of depth. (Google, 2026), (Nagoor, 2026)
- **Narrative Text Integration:** A major focus was the integration of dynamic bitmap text using `glutBitmapCharacter` to drive the storytelling. We developed a custom `drawCenteredText` utility that calculates string widths to ensure perfect alignment for scene titles, interactive prompts, and the final philosophical message. This allowed us to provide clear user guidance and emotional weight to the "Fragments of Growth" experience. (The Khronos Group, n.d.-a)



2.2 Libraries Used:

The project was developed in C++ using the OpenGL framework, leveraging a suite of libraries to handle everything from low-level rendering to high-level user interaction:

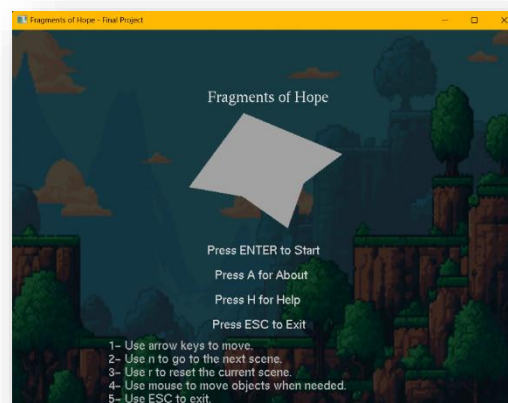
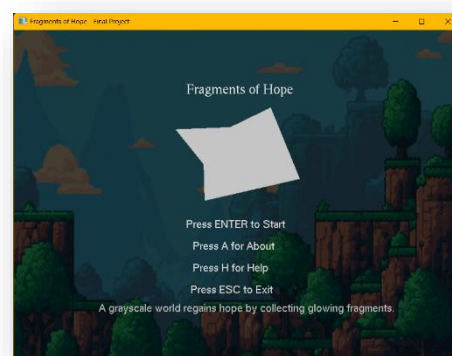
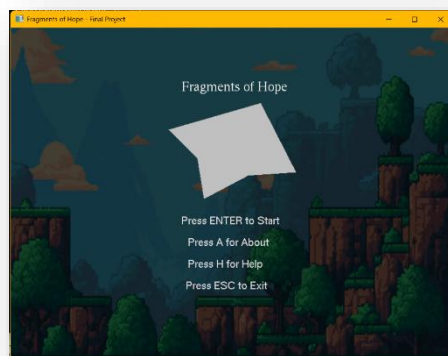
- **FreeGLUT:** Utilized as the primary toolkit for window creation, managing the display lifecycle, and handling essential input callbacks for the keyboard and mouse.
- **GLEW (OpenGL Extension Wrangler):** Employed to manage OpenGL extensions and modern entry points, ensuring cross-platform compatibility and access to advanced rendering features.
- **Math.h:** Critical for geometric computations, specifically for calculating Euclidean distance using the Pythagorean theorem to facilitate precise collision detection between the player and fragments.
- **Stdlib.h & String.h:** Used for fundamental memory management, such as allocating buffers for texture data, and for manipulating C-style strings used in file path handling and text rendering.
- **iostream:** Integrated for standard input/output operations, primarily for debugging purposes and providing console-based feedback during texture loading and state transitions.
- **Vector & String (C++ Standard Library):** Utilized for modern data management, allowing for the organized storage of game objects, scene metadata, and dynamic text strings.



Chapter 3: Results

3.1 Scene 1: The Narrative Gateway (Cover Scene)

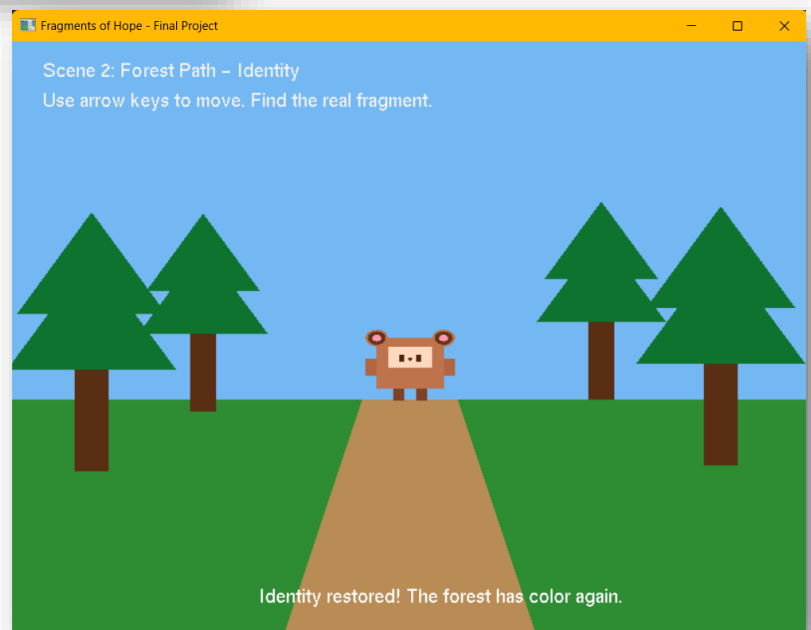
The opening scene establishes the atmosphere of the project. It features a high-resolution texture background loaded via a custom BMP utility. Central to this scene is a pulsing, rotating fragment that utilizes **glRotatef** and **glScalef** to draw the user's attention. A text-based UI provides immediate guidance on controls, including "About" and "Help" toggles.





3.2 Scene 2: Forest Path (Identity)

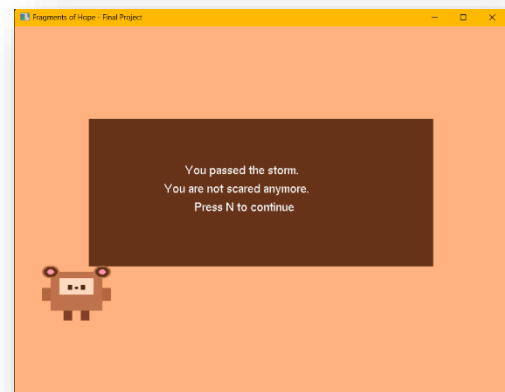
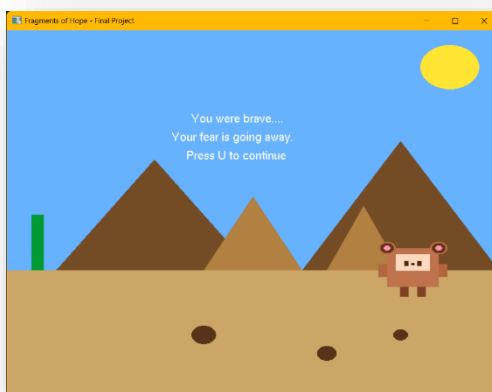
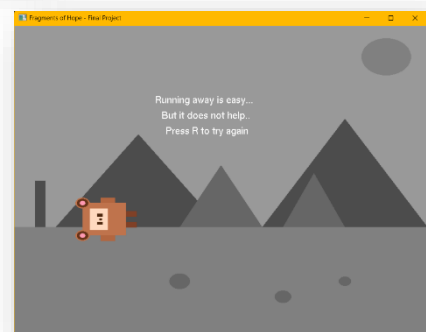
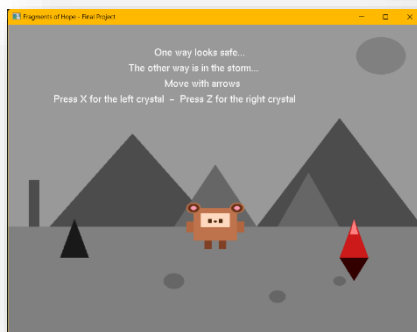
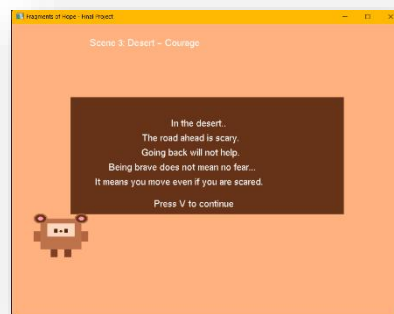
In this scene, the player must identify the "real" fragment among decoys. Upon collision—calculated via a Euclidean distance-check algorithm—the `identitySolved` boolean triggers a dramatic color shift in the background from a desaturated gray to vibrant blues and greens, simulating the restoration of the world's identity.





3.3 Scene 3: The Desert (Courage)

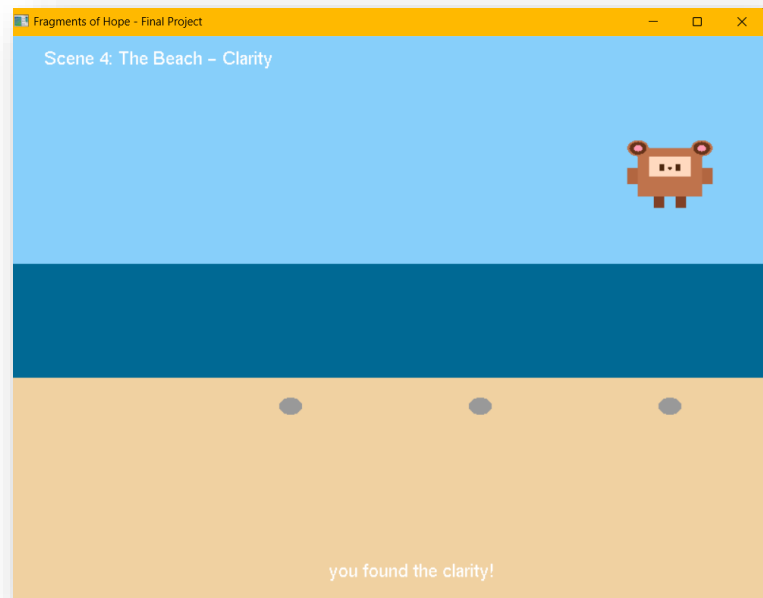
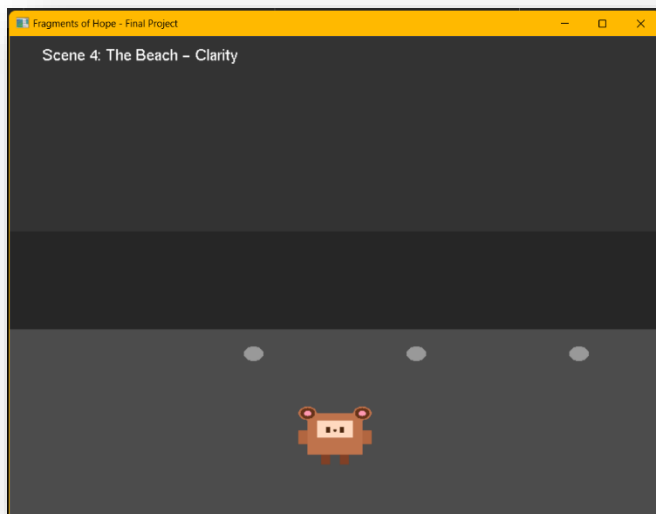
This scene is divided into narrative and gameplay phases. The player navigates a harsh desert environment featuring mountains and cacti rendered with `GL_TRIANGLES` and `GL_QUADS`. To progress, the player must demonstrate "Courage" by choosing a path through a storm. A "Lose" state was implemented using a **rotation transformation (fallAngle)** to animate the character falling if the incorrect choice is made.





3.4 Scene 4: The Serene Beach (Clarity)

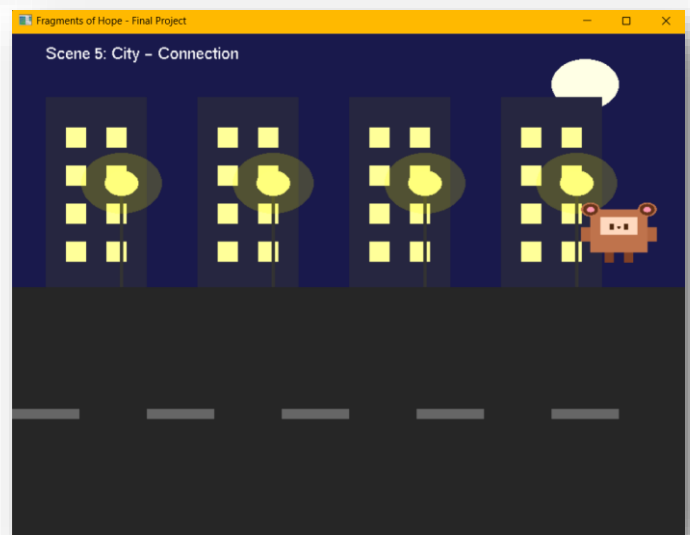
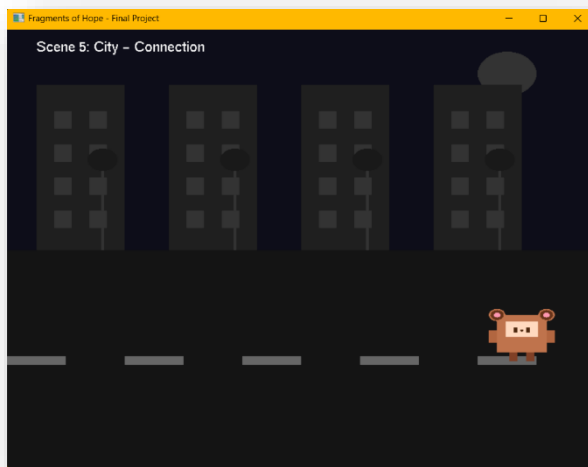
The Beach scene introduces a "Sea Barrier" logic where the player must find a specific path to achieve "Clarity." If the player enters the water without solving the logic, their position is reset. Once the **win-zone** is reached, the grayscale sand and water transition into bright, saturated colors, and a hidden fragment appears for collection.





3.5 Scene 5: The Urban City (Connection)

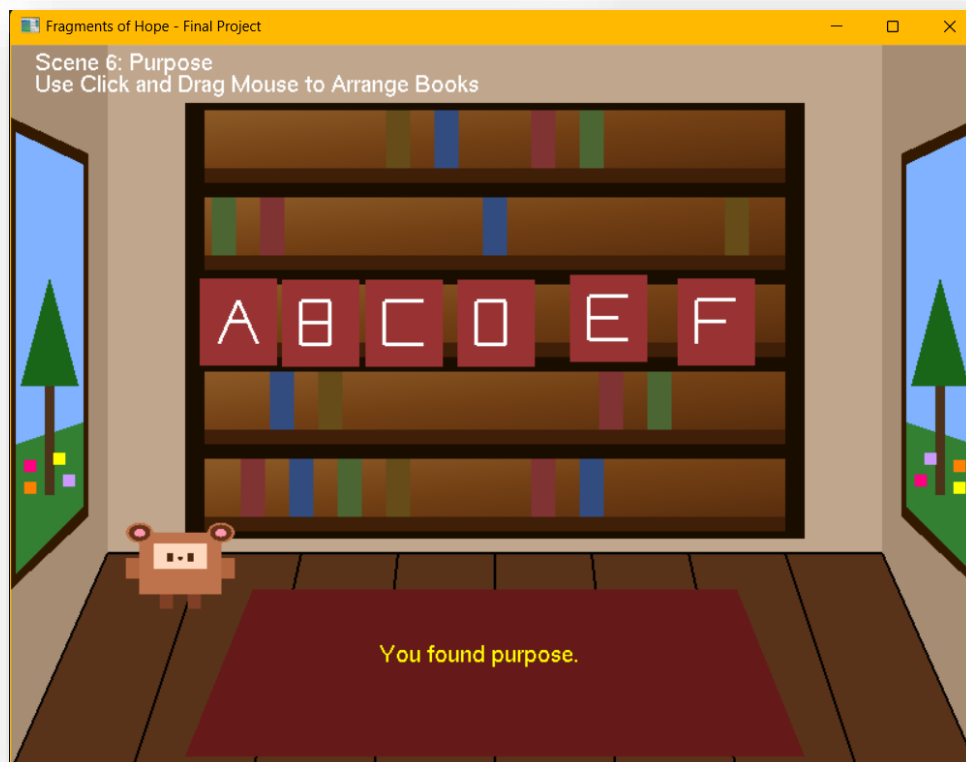
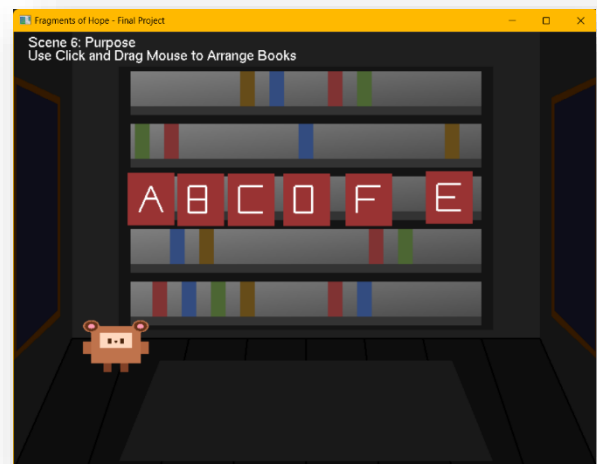
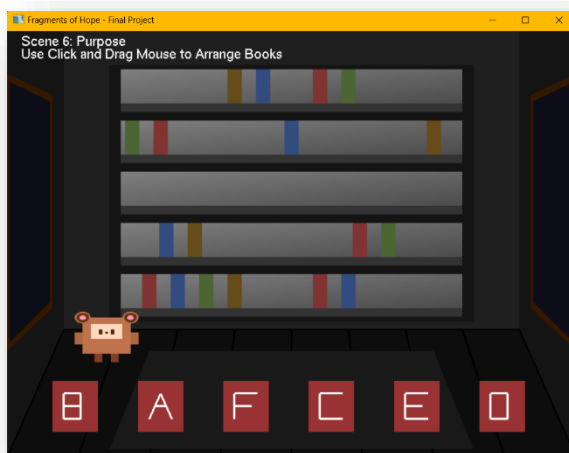
The city scene focuses on a sequential puzzle. The player must move the character to touch four "Nodes" in a specific order. Each successful touch changes the node's color to yellow. Once all nodes are connected, the city's dark, monolithic buildings and sky shift into a bright, connected environment, representing the restoration of social bonds.





3.6 Scene 6: The Library (Purpose)

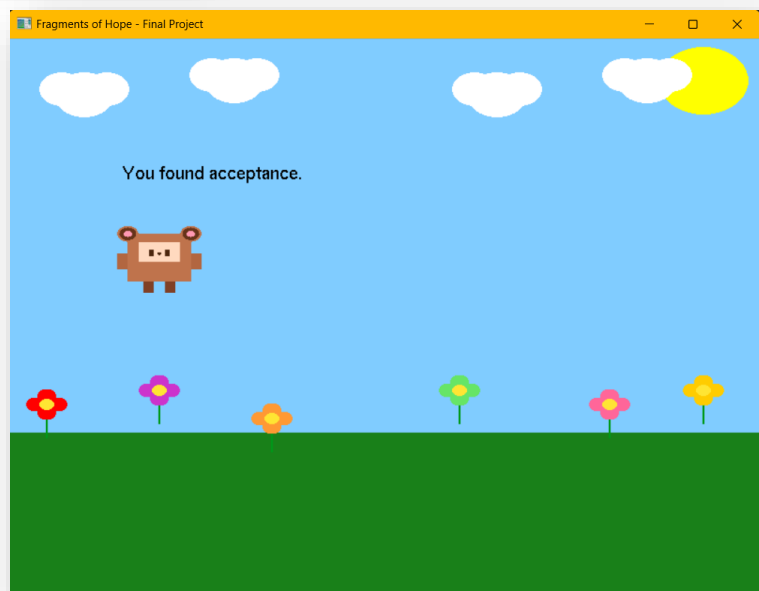
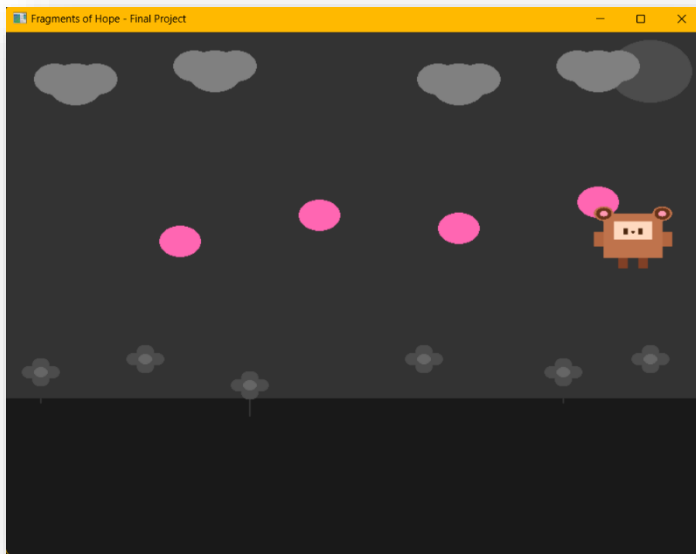
The library serves as the project's most complex interaction scene. Using **mouse coordinates** converted to OpenGL world space, **players must click and drag books** to sort them in the **correct alphabetical order**. This scene demonstrates the successful integration of mouse picking, real-time object translation, and 3D depth management into a 2D narrative.





3.7 Scene 7: The Garden (Acceptance)

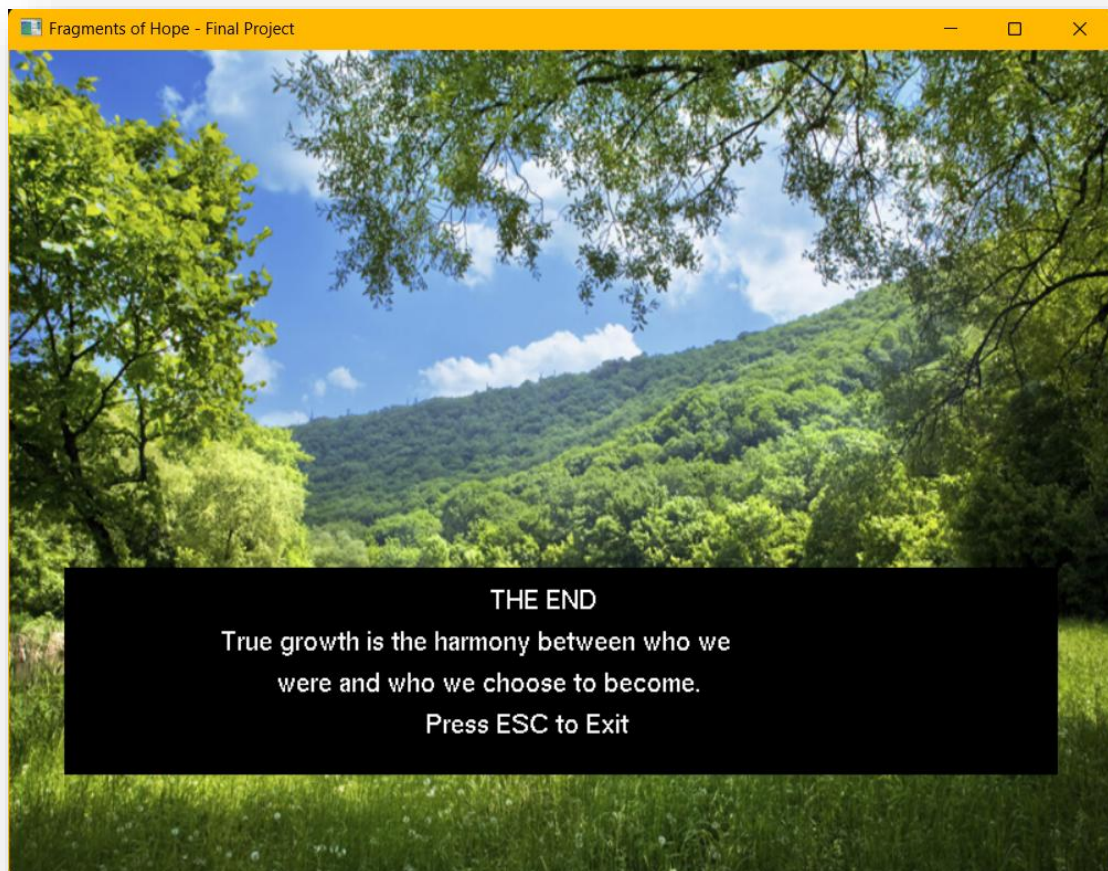
In the penultimate scene, the player must collect four final fragments scattered across a garden. As each fragment is collected, the grayscale flowers and clouds regain their natural colors. This scene utilizes `glClearColor` updates to transition the entire world's atmosphere once the `scene7Solved` condition is met.





3.8 Scene 8: Final Light (Ending)

The project concludes with a final texture-mapped screen and a centered narrative message. This scene serves as a reflection on the journey, utilizing `drawCenteredText` to display a final thematic statement. The user is then given the option to exit the application gracefully using the **ESC key**.





Technical Reflection & Collaborative Analysis

Reflection on Lessons Learned

The development of **Fragments of Hope** provided a rigorous test of our understanding of the **OpenGL state machine**. A primary technical challenge was ensuring that geometric transformations—specifically `glRotatef` and `glScalef`—remained isolated to their intended objects. We overcame this by implementing strict `glPushMatrix` and `glPopMatrix` protocols within our display functions.

Furthermore, we had to manage the complexity of integrating 2D UI elements with the **3D library scene** and **textures**. This required us to strategically use `glEnable` and `glDisable` for `GL_TEXTURE_2D` and `GL_DEPTH_TEST` multiple times across the project. By carefully toggling these states, we ensured that 2D overlays **remained visible** and that textures did not interfere with the rendering of primitive shapes in other scenes. Additionally, managing the **User Interaction Layer** required a deep dive into converting screen-space mouse coordinates into world-space coordinates, a crucial step for the alphabetical book-sorting puzzle in Scene 6. This experience bridged the gap between mathematical theory and interactive graphical applications.

Group Collaboration & Effectiveness

To **work effectively as a group**, we adopted a modular development strategy. By dividing the project into six distinct scenes, each member focused on specific implementation logic—such as the mouse-picking system or the garden collection logic—while maintaining a unified code structure. We held regular synchronization sessions to ensure that global state variables, such as `currentScene` and logic flags like `identitySolved`, functioned seamlessly across the entire pipeline. This collaborative approach allowed us to maintain high-quality control standards and meet our internal deadlines well ahead of the final submission.

Future Work

While the current version of the project successfully implements all core requirements, we identified several areas for technical expansion:

- **Spatial Audio Integration:** We plan to implement a dedicated audio engine to provide immersive, auditory feedback that responds to the player's proximity to fragments.
- **Advanced Shading:** Moving beyond basic color-blending, the project could be enhanced using GLSL shaders to implement real-time lighting and shadow mapping for increased depth.



References

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- The Khronos Group. (n.d.). *FreeGLUT: The open-source OpenGL utility toolkit*. OpenGL.org. [FreeGLUT](#)
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- *Plagiarism and AI (How to properly cite AI - APA 7th & UP standards)*. (2026, May 12). [Video]. YouTube. Retrieved May 14, 2026, from [Plagiarism and AI](#) To